

Equation 7 Derivation

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Assumptions

$$\begin{aligned} g_{t+1} &= g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, I), \quad \Sigma_{g,r} = \sigma_{g,r} \sigma_{g,r}^\top = \text{diag}(\sigma_{g1,r}^2, \dots, \sigma_{gq_r,r}^2). \\ \log M_{t+1} &= -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1}, \quad \varsigma_r^2 = \lambda_r^\top \Sigma_{g,r} \lambda_r + \sigma_v^2, \quad v_{t+1} \sim \mathcal{N}(0, \sigma_v^2). \\ X_{i,t} &= \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top z_t + u_{i,t}, \quad S_{i,t} := \Gamma_i^\top z_t + u_{i,t}, \quad \Delta S_{i,t} \text{ stationary with } \mathbb{E}_t[\Delta S_{i,t}] = 0. \end{aligned}$$

No Arbitrage

$$1 = \mathbb{E}_t[M_{t+1} e^{X_{i,t} - X_{i,t+1}}], \quad Y_{t+1} := \log M_{t+1} + X_{i,t} - X_{i,t+1}.$$

Computing $X_{i,t} - X_{i,t+1}$

Write the affine form at t and $t+1$

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + S_{i,t}, \quad X_{i,t+1} = \mu_{i,r} + \lambda_{i,r}^\top g_{t+1} + S_{i,t+1},$$

where $S_{i,t} := \Gamma_i^\top z_t + u_{i,t}$.

Subtract the two expressions

$$\begin{aligned} X_{i,t} - X_{i,t+1} &= (\mu_{i,r} + \lambda_{i,r}^\top g_t + S_{i,t}) - (\mu_{i,r} + \lambda_{i,r}^\top g_{t+1} + S_{i,t+1}) \\ &= \lambda_{i,r}^\top (g_t - g_{t+1}) - (S_{i,t+1} - S_{i,t}) \quad (\text{the two } \mu_{i,r} \text{ cancel}) \\ &= -\lambda_{i,r}^\top (g_{t+1} - g_t) - \Delta S_{i,t}, \end{aligned}$$

where $\Delta S_{i,t} := S_{i,t+1} - S_{i,t}$.

Expand $\Delta S_{i,t}$

$$\Delta S_{i,t} = \Gamma_i^\top (z_{t+1} - z_t) + (u_{i,t+1} - u_{i,t}) = \Gamma_i^\top \Delta z_t + \Delta u_{i,t}.$$

Use the paper's factor law for g_{t+1}

$$g_{t+1} - g_t = \mu_r + \sigma_{g,r} \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, I).$$

$$\begin{aligned} X_{i,t} - X_{i,t+1} &= -\lambda_{i,r}^\top (\mu_r + \sigma_{g,r} \varepsilon_{t+1}) - \Delta S_{i,t} \\ &= -\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta S_{i,t}. \end{aligned}$$

Assemble Y_{t+1}

Define the object whose expectation we evaluate

By the FX restriction $1 = \mathbb{E}_t[M_{t+1}e^{X_{i,t}-X_{i,t+1}}]$, set

$$Y_{t+1} := \log M_{t+1} + X_{i,t} - X_{i,t+1}.$$

Substitute the SDF and the difference from the previous subsection

Using $\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1}$ and $X_{i,t} - X_{i,t+1} = -\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta S_{i,t}$, we get

$$\begin{aligned} Y_{t+1} &= \left(-r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1} \right) + \left(-\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta S_{i,t} \right) \\ &= -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_{t+1} - \lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}. \end{aligned}$$

Expand g_{t+1} using the factor law

Since $g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}$,

$$-\lambda_r^\top g_{t+1} = -\lambda_r^\top (g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}) = -\lambda_r^\top g_t - \lambda_r^\top \mu_r - \lambda_r^\top \sigma_{g,r} \varepsilon_{t+1}.$$

Substitute and collect like terms

$$\begin{aligned} Y_{t+1} &= -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - \lambda_r^\top \mu_r - \lambda_r^\top \sigma_{g,r} \varepsilon_{t+1} \\ &\quad - \lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t} \\ &= -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r - (\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}. \end{aligned}$$

Identify deterministic vs. shock terms

$$\underbrace{-r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r}_{\text{deterministic/state part}} + \underbrace{-(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}}_{\text{innovations}}.$$

Mean/variance and log-normal expectation

$$\begin{aligned} \mathbb{E}_t[Y_{t+1}] &= -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r, \\ \text{Var}_t(Y_{t+1}) &= (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}). \end{aligned}$$

Compute $\text{Var}_t(Y_{t+1})$

Start from the assembled Y_{t+1}

From the previous subsection we have

$$Y_{t+1} = \underbrace{-r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r}_{\text{deterministic, } \mathcal{F}_t\text{-measurable}} - \underbrace{(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}}_{(a)} - \underbrace{v_{t+1}}_{(b)} - \underbrace{\Delta S_{i,t}}_{(c)}.$$

Identify the random innovations and their properties

By assumption in regime r :

$$\varepsilon_{t+1} \sim \mathcal{N}(0, I), \quad v_{t+1} \sim \mathcal{N}(0, \sigma_v^2), \quad \Delta S_{i,t} \text{ is stationary with finite variance.}$$

Moreover, ε_{t+1} is *independent* of v_{t+1} and of the stationary block, and all have zero conditional mean given $\mathcal{F}_t$. Also $\Sigma_{g,r} = \sigma_{g,r} \sigma_{g,r}^\top = \text{diag}(\sigma_{g1,r}^2, \dots, \sigma_{gq,r}^2)$.

Drop the deterministic part for the variance

Let

$$A_t := -r_t - \frac{1}{2}\zeta_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r \quad (\mathcal{F}_t\text{-measurable}),$$

and write

$$Y_{t+1} = A_t - (\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}.$$

Since A_t is $\mathcal{F}_t$ -measurable, $\text{Var}_t(A_t) = 0$. Hence

$$\text{Var}_t(Y_{t+1}) = \text{Var}_t\left(-(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta S_{i,t}\right).$$

Use variance-of-sum with independence

Because ε_{t+1} , v_{t+1} , and $\Delta S_{i,t}$ are mutually independent (and each has conditional mean zero),

$$\text{Var}_t(Y_{t+1}) = \text{Var}_t((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}) + \text{Var}_t(v_{t+1}) + \text{Var}_t(\Delta S_{i,t}).$$

Recall, for any scalar c and random variable X ,

$$\text{Var}(cX) = c^2 \text{Var}(X),$$

So $\text{Var}(x) = \text{Var}(-x)$.

Important Variance Property

The property that $\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z)$ for independent random variables is fundamental in probability theory and statistics.

General Variance Formula

For any random variables X , Y , and Z , the general formula for the variance of their sum is:

$$\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z) + 2\text{Cov}(X, Y) + 2\text{Cov}(X, Z) + 2\text{Cov}(Y, Z)$$

Why Independence Eliminates Covariance Terms

Independence means that knowing the value of one variable gives no information about another. Mathematically, this translates to:

- If X and Y are independent, then $E[XY] = E[X]E[Y]$
- Covariance is defined as: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
- Therefore, for independent X and Y :

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0$$

Mathematical Proof

Starting with the definition of variance:

$$\text{Var}(X + Y + Z) = E[(X + Y + Z)^2] - [E(X + Y + Z)]^2$$

When we expand $(X + Y + Z)^2$, we get:

$$(X + Y + Z)^2 = X^2 + Y^2 + Z^2 + 2XY + 2XZ + 2YZ$$

Taking expectations:

$$E[(X + Y + Z)^2] = E[X^2] + E[Y^2] + E[Z^2] + 2E[XY] + 2E[XZ] + 2E[YZ]$$

For the second term:

$$\begin{aligned} [E(X + Y + Z)]^2 &= [E[X] + E[Y] + E[Z]]^2 \\ &= E[X]^2 + E[Y]^2 + E[Z]^2 + 2E[X]E[Y] + 2E[X]E[Z] + 2E[Y]E[Z] \end{aligned}$$

Subtracting equation (7) from equation (6):

$$\begin{aligned} \text{Var}(X + Y + Z) &= E[X^2] + E[Y^2] + E[Z^2] + 2E[XY] + 2E[XZ] + 2E[YZ] \\ &\quad - E[X]^2 - E[Y]^2 - E[Z]^2 - 2E[X]E[Y] - 2E[X]E[Z] - 2E[Y]E[Z] \\ &= [E[X^2] - E[X]^2] + [E[Y^2] - E[Y]^2] + [E[Z^2] - E[Z]^2] \\ &\quad + 2[E[XY] - E[X]E[Y]] + 2[E[XZ] - E[X]E[Z]] + 2[E[YZ] - E[Y]E[Z]] \\ &= \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z) + 2\text{Cov}(X, Y) + 2\text{Cov}(X, Z) + 2\text{Cov}(Y, Z) \end{aligned}$$

Under independence, all covariance terms equal zero, giving us:

$$\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z)$$

Compute each variance term

$$\begin{aligned} \text{Var}_t((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}) &= (\lambda_{i,r} + \lambda_r)^\top \underbrace{\sigma_{g,r} \text{Var}_t(\varepsilon_{t+1}) \sigma_{g,r}^\top}_{\sigma_{g,r} I \sigma_{g,r}^\top = \Sigma_{g,r}} (\lambda_{i,r} + \lambda_r) \\ &= (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r), \\ \text{Var}_t(v_{t+1}) &= \sigma_v^2, \\ \text{Var}_t(\Delta S_{i,t}) &= \text{Var}_t(\Delta(\Gamma_i^\top z_t + u_{i,t})) \quad (\text{finite by assumption}). \end{aligned}$$

Variance of the global-factor innovation term

We want

$$\text{Var}_t((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}),$$

where $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$ is a $q_r \times 1$ vector of i.i.d. standard normals, and $\sigma_{g,r}$ is a $q_r \times q_r$ diagonal matrix with entries $\sigma_{g1,r}, \dots, \sigma_{gq_r,r}$.

Factor out the deterministic coefficients

Since $(\lambda_{i,r} + \lambda_r)$ and $\sigma_{g,r}$ are both deterministic conditional on $\mathcal{F}_t$,

$$\text{Var}_t((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}) = (\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \text{Var}_t(\varepsilon_{t+1}) \sigma_{g,r}^\top (\lambda_{i,r} + \lambda_r).$$

Recall, for any vector a and random vector X ,

$$\text{Var}(a^\top X) = a^\top \text{Var}(X) a.$$

Use the variance of the standard normal vector

By assumption, $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$, so

$$\text{Var}_t(\varepsilon_{t+1}) = I.$$

Simplify the middle term

$$\sigma_{g,r} \text{Var}_t(\varepsilon_{t+1}) \sigma_{g,r}^\top = \sigma_{g,r} I \sigma_{g,r}^\top = \sigma_{g,r} \sigma_{g,r}^\top.$$

Recognize the definition of $\Sigma_{g,r}$

By the paper's definition,

$$\Sigma_{g,r} := \text{diag}(\sigma_{g1,r}^2, \dots, \sigma_{gq_r,r}^2) = \sigma_{g,r} \sigma_{g,r}^\top.$$

Substitute back

Therefore,

$$\text{Var}_t\left((\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1}\right) = (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r).$$

Combining the pieces

$$\text{Var}_t(Y_{t+1}) = (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}).$$

Since $Y_{t+1} | \mathcal{F}_t$ is normal, $\mathbb{E}_t[e^{Y_{t+1}}] = \exp(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1})) = 1$, so

$$\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1}) = 0.$$

Theoretical Clarification

The phrase

$$Y_{t+1} | \mathcal{F}_t$$

means that we are considering the conditional distribution of the random variable Y_{t+1} given the information set (filtration) $\mathcal{F}_t$, which contains all information available up to time t . Saying that

$$Y_{t+1} | \mathcal{F}_t \sim \mathcal{N}(\cdot, \cdot)$$

means that, conditional on what is known at time t , the distribution of Y_{t+1} is normal (Gaussian). This assumption follows from the model setup: the shocks $\varepsilon_{t+1}, v_{t+1}, \Delta S_{i,t}$ are Gaussian and independent, so any linear combination of them, such as Y_{t+1} , is also Gaussian. Because $Y_{t+1} | \mathcal{F}_t$ is normal, we can apply the log-normal expectation identity:

$$\mathbb{E}_t[e^{Y_{t+1}}] = \exp(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1})).$$

Plugging in the conditional mean and the conditional variance

From above,

$$\mathbb{E}_t[Y_{t+1}] = -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r.$$

Remember, we derived

$$\text{Var}_t(Y_{t+1}) = (\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}).$$

So,

$$0 = \mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1}) = -r_t - \frac{1}{2} \varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2} \left[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}) \right].$$

$$C_t := -r_t - \frac{1}{2} \varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2} \left[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta S_{i,t}) \right].$$

$$\boxed{\mu_{i,r} := [C_t | r] = -\bar{r}_r - \frac{1}{2} \varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2} \left[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r} (\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta(\Gamma_i^\top z_t + u_{i,t})) \right].}$$

Affine form and why the shock $S_{i,t}$ drops out of the mean

We assume the affine decomposition

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + S_{i,t},$$

where $\mu_{i,r}$ is a constant (intercept), $\lambda_{i,r}^\top g_t$ is the systematic state term, and $S_{i,t}$ is an idiosyncratic shock with zero conditional mean:

$$\mathbb{E}[S_{i,t}] = 0.$$

If $Y = \log(X_{i,t})$, then

$$\mathbb{E}[e^Y] = \mathbb{E}[X_{i,t}].$$

Taking conditional expectation of the affine form gives

$$\mathbb{E}[X_{i,t}] = \mathbb{E}[\mu_{i,r}] + \mathbb{E}[\lambda_{i,r}^\top g_t] + \mathbb{E}[S_{i,t}] = \mu_{i,r} + \lambda_{i,r}^\top g_t + 0,$$

so the random shock $S_{i,t}$ drops out and the mean of $X_{i,t}$ is just the deterministic affine part:

$$\mathbb{E}[X_{i,t}] = \mu_{i,r} + \lambda_{i,r}^\top g_t.$$

That is why, once we factor out the g_t contribution, the remaining term is simply the constant $\mu_{i,r}$.